



Lightweight Convolutional Neural Network for Visual Image Reconstruction from fMRI Using Limited Data

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Abstract. The majority of the latest models of visual reconstruction based on fMRI are computationally demanding, which renders this study unaffordable to others. Instead, we chose to tackle this trend by emphasizing a convenient and efficient solution. Our work presents a lightweight U-Net, which is used to reconstruct pictures based on the Generic Object Decoding dataset. In order to address the issue of scarce data, we constructed an effective pipeline with the help of Principal Component Analysis (PCA) to reduce the number of features, as well as robust regularization and Mix-up augmentation. The held-out test set also generalized well to our model, with images that showed apparent structural similarity and with a significant score of SSIM of approximately 0.24. This study proves that it is entirely possible to perform brain decoding resource-efficiently and, more importantly, it sets a crucial performance-cost point of reference in a broad field.

Keywords: fMRI Image Reconstruction · Lightweight CNN · Brain Decoding · Computational Efficiency · Limited Data · U-Net

1 Introduction

The neuroscience of being able to see what one sees through just a scan of the brain is one of the most exciting achievements in the field. Through fMRI, we are able to have an in-depth, non-invasive view of what is going on in the brain. Although simple patterns were the most nascent work a human could identify, current state-of-the-art deep learning models can synthesize images of unexpectedly good fidelity. As state-of-the-art models are expensive in computation, this development requires a huge cost; they need to use a large cluster of GPUs, and it takes days or even weeks of training. This makes it nearly an insurmountable barrier to most researchers and poses basic questions about whether these models are merely memorizing those small, noisy fMRI datasets that they are trained on. This led us to ask ourselves: Is this only the way: bigger is better? Or is there a more innovative, efficient, and lightweight model that can, nonetheless, do the job? In response to this, we devised a comparative study. We created a simple encoder-decoder-based Baseline model and provided it with the raw and high-dimensional fMRI data, and asked whether it would learn on its own. Then we tried an Advanced hybrid

model. The fMRI data have been cleaned with the help of PCA, and the most important signals are extracted. And the detail-preserving U-Net architecture has then been trained on the optimized signal using effective regularization, such as Mix-up, to make it learn real patterns, not noise. Through comparison of these two approaches, we get a definite answer in the trade-off between performance and cost. We show that even a lightweight and user-friendly pipeline can obtain significant results within minutes, which is a crucial step and makes the field ask itself why its larger-than-life models can cost that much.

2 Related Works

Our method, with the difficulty of unraveling the brain activity with the help of computer vision and specific neuroscience programs, has the benefit of computational efficiency and does not require a lack of data. In the center of the structural architecture is U-Net [1], which includes encoder-decoder, skip connection, but the state-of-the-art results are mostly given by generative networks. GANs are the best models to synthesize photo-realistic images [2], and for learning structured latent spaces in high-dimensional data, VAEs are best [5]. Pixel-level error and human perception's link happen through perceptual loss functions for image optimization in the feature space [3]. This knowledge enables us to generate the quality of perception at a low cost comparatively a full generative model. fMRI decoding carries this technique, but the brain and image are brought closer together by DNNs [15], and VAEs and GANs [9, 18], MinD-Vis are combined as the state-of-the-art models by GANs [4, 8], so the task has been broken down into steps. However, in the literature, simple linear models can also be determined from natural images of brain activity, which suggests that brute force complexity is not the only way for meaningful results. We use the U-Net architecture for fMRI-to-image mapping and demonstrate the capability to reconstruct fMRI-to-image of a specific class [13], by DNNs [11], and feature-based optimization happens and support comes for our perceptual loss-based approach from Brain Reading Studies Cox and Savoy [20]. Bigger models have certain challenges in terms of efficiency and sparsity. Nevertheless, in the medical imaging community, the medical imaging deep learning-based approach to MRI reconstruction acceleration and compressed sensing is the way out to efficient models [7, 10]. The fMRI data lowering of the research lies in the self-supervised learning research to the few-shot learning research [6]. In exchange for VAE-GAN quality [9], we have a light U-Net with strong regularization, Mix-up [21], and feature-based perceptual loss to generalize over overfitting [12].

3 Dataset

We have used the Generic Object Decoding (GOD) dataset [17]. We have focused on the data collected from Subject 1, which contains 1200 training images and 50 testing images in the corresponding stimulus file [19]. The body package has been used for loading and data handling and is stored in the Horikawa-Kamitani Lab in Bdata format in HDF5 files. Focusing on the visual cortex, we get relevant voxels [14].

4 Methodology

We possess a two-phase methodology. They are:

1. Baseline Method (Direct Reconstruction)
2. Advanced Method (Preprocessing-Informed Reconstruction).

The conventional data pre-processing, model training and evaluation, feature extraction, and model design procedures have been adhered to (Fig. 1).

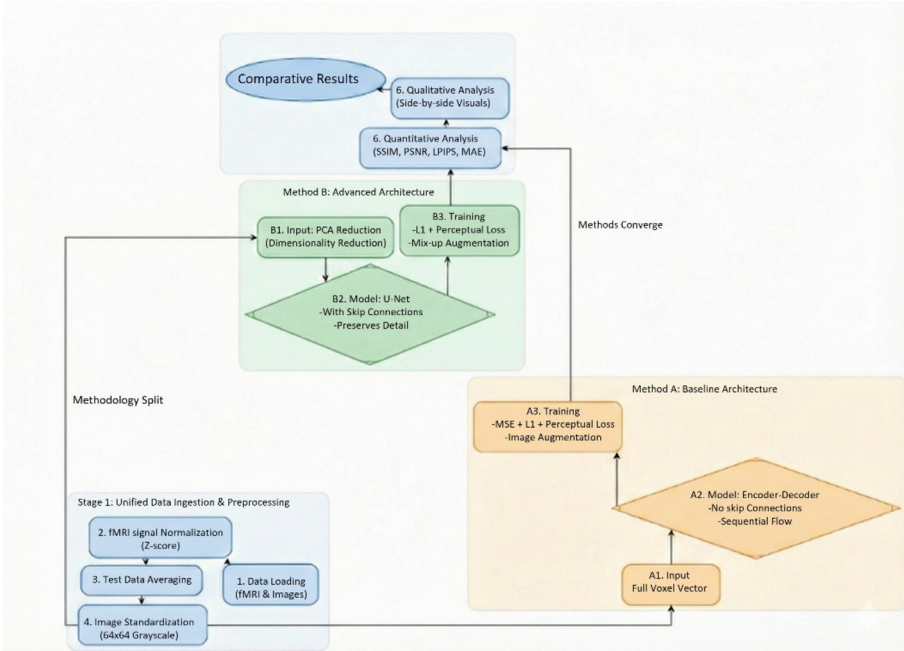


Fig. 1. A comparative methodology of fMRI-based image reconstruction, comprising baseline architecture and advanced architecture of evaluation and comparison.

4.1 Unified Data Ingestion and Initial Preprocessing

This initial stage is

1. **Data Loading:** fMRI dataset (in the form of .h5 files) and stimulus images are loaded with fMRI signals of the visual cortex (ROI_VC), and also with the links of stimulus that are related to each scan are stored in the dataset.
2. **fMRI Signal Normalization:** We standardize the raw voxels to ensure stable training. The training data and the test data were normalized through Z-score, which was calculated as $(fMRI_{norm} = fMRI_{raw} - \mu/\sigma)$ by using the standard deviation and the mean of the training data.

3. **Test Data Smoothing:** Several (35) fMRI scans over the same test images have been averaged to obtain cleaner test data.
4. **Image Standardization:** We transform and resize the stimulus images into grayscale along with the same resolution—e.g., 64×64 pixels and then scale their pixel values within the range $[0.0, 1.0]$.

4.2 Architectural Branches

We split the architectural branch into two parallel paths; each contains a unique approach to feature engineering and model architecture.

Method A: Baseline Encoder-Decoder Architecture

Here, we create a direct, feed-forward approach to test the hypothesis that a lightweight CNN can overcome the curse of dimensionality in raw fMRI data directly through its own feature learning for translating brain activity into an image.

- **A1. Preprocessing:** We directly use the full, high-dimensional fMRI vector (e.g., 3444 features) from Stage 1. We don't do any dimensionality reduction.
- **A2. Model Architecture (Simple Encoder-Decoder):**
 - **fMRI Encoder:** Full fMRI vectors are compressed into a low-dimensional latent representation by MLP.
 - **Image Decoder:** Compressed representations are achieved by a series of deconvolution layers and progressive up-sampling for the final 64×64 reconstruction.
 - **Key Characteristic:** In the absence of any skipping connections and fine-grained detail loss, the network sequentially flows the information.
- **A3. Training and Regularization:**
 - **Loss Function:** The combination of MSE, L1, and Perceptual (VGG-16 with frozen weights) Loss gives more accurate perceptual loss as a static, off-the-shelf tool.
 - **Augmentation:** Image-space augmentations (rotation, jitter) are applied to the training images.
 - **Scheduler:** Adjustment appears through Basic StepLR in the Learning rate scheduler.

Omitting parameter-heavy components like U-Net skip connections and GAN discriminators, the minimalist three-layer decoder makes this model lightweight. Low parameter count ($\sim 11.26\text{M}$) directly results in simplicity in architectural status, making it a computationally efficient baseline.

Method B: Advanced U-Net with PCA

More advanced feature reduction techniques and a stronger image generation model have been used to test our main hypothesis: that for a lightweight model to succeed, an intelligent preprocessing step is essential to denoise and compress the signal before it is fed to a more powerful architecture.

- **B1. Preprocessing (PCA with Dimensionality Reduction):**

- Before presenting the model, Principal Component Analysis is used to preprocess the normalized fMRI data.
- Reducing the feature space, the complete voxel count comes to a smaller number of principal components (e.g., 1000), which captures the most variance in the data, denoises, and selects essential features.

- **B2. Model Architecture (True U-Net):**

- **fMRI Encoder Head:** The reduced PCA vectors are projected into an initial feature map by an MLP.
- **U-Net Body:** An encoder-decoder (contraction-expansion) exists in the core of the U-Net
- **Key Characteristic:** The skip connection between the expanding path's layers with the corresponding layers of the contracting path enables the model's retention of high-resolution spatial information, essential for accurate, clear images.

- **B3. Training and Regularization:**

- **Loss Function:** L1 and Perceptual (VGG-16 with frozen weights) combined in a focused loss.
- **Augmentation:** In advanced Mix-up augmentation, new training samples have been created in the linear combination of existing fMRI-image pairs.
- **Scheduler:** A more sophisticated CosineAnnealingLR scheduler is used for smoother learning rate decay.

The lightweight nature of the Advanced model originates from a hybrid approach that combines PCA-based fMRI vector reduction to 1000 components with a reduced U-Net architecture. This leads to decreasing parameters in the early layers. Containing skip connections, the shallow U-Net contains two down-sampling levels and limited channel depth to preserve details without the larger model's extravagance. Rendering efficiently and capably, the parameter size increases to approximately 17.97M.

Stage 3: Unified Evaluation and Comparison

After training, we evaluate both models on the test set using standardized metrics to definitively compare the outcomes of our two hypotheses and determine the best method for resource-efficient brain decoding.

Quantitative Analysis: Performance is measured using a suite of metrics:

SSIM (Structural Similarity Index Measure): Offering a quality assessment in contrast to PSNR or MAE, it evaluates image similarity by examining luminance, contrast, and structure.

For two image windows, x and y :

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

where:

- μ_x and μ_y show the average luminance values.
- σ_x^2 and σ_y^2 denote the variances (contrast).
- σ_{xy} is the covariance (structural similarity).
- c_1 and c_2 denote constants, used to guarantee the stability of the division.

A SSIM score of 1 represents a structural similarity. Though it ranges from 0 to 1 as usual but it's in $[-1, 1]$. A high score is better.

PSNR (Peak Signal-to-Noise Ratio): In terms of the reconstruction error (MSE) in dB, it is the power ratio which's higher value gives better image quality.

The calculation of PSNR is as follows:

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

where:

- MAX_I shows the pixel intensity value (for example, 255, in 8-bit images). So, PSNR greater than or equal to 30 represents better image quality.

LPIPS (Learned Perceptual Image Patch Similarity): LPIPS is a measurement algorithm for image similarity from the analysis of feature maps, which comes from a trained neural network for quantifying perceptual variation. The conceptual formula is:

$$d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{n,w} ||w_l \odot f_w^l - f_{0hw}^l||_2^2$$

Put plainly, this signifies:

- For each layer l in the network, extract the feature maps f^l and f_0^l Associated with the two images.
- Gives the pixel-wise difference between them.
- learned weights w_l scales this difference.
- Average the results in each layer and sum them all.

LPIPS measures the distinction between a pair of images. A smaller score means the images are more similar. A value of 0 indicates they are identical. Smaller scores are better.

MAE (Mean Absolute Error): The discrepancy between the original and the reconstructed image's pixels is given by MAE. Due to the equal consideration of all pixels, an accurate reflection of the assessment of image quality may not be obtained.

For two images, an original y and a reconstruction $\hat{y}$, each with n pixels:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where:

- y_i represents the original image's i -th pixel.
- $\hat{y}_i$ denotes the reconstructed image's i -th pixel.
- n_i denotes the total number of pixels.

The score represents the average pixel error. A score of 0 means the images are identical. Lower is better.

Qualitative Analysis: A side-by-side visualization with the original ground truth images for the best reconstructions from both methods is given for the direct human assessment of visual fidelity.

5 Results

Baseline (no PCA) and Advanced (with PCA) methodologies have been compared on the Generic Object Decoding (GOD) dataset that includes quantitative image reconstruction metrics, visual comparisons, and an analysis of dynamic training and efficient trade-offs.

5.1 Quantitative Analysis

Based on the Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), Learned Perceptual Image Patch Similarity (LPIPS), and Mean Absolute Error (MAE), a quantitative performance evaluation was done on the held-out test set. Table 1 shows a stark contrast in both models' efficient approach.

A complete performance breakdown for both models can be seen in Table 1.

Table 1. Quantitative performance and efficiency comparison.

Model	SSIM score (higher is better)	PSNR (dB) (higher is better)	LPIPS score (lower is better)	MAE (lower is better)	Model parameters
Baseline (without PCA)	0.2425	11.2559	0.6208	0.2307	~ 11.26M
Advanced (with PCA)	0.2438	11.2659	0.6108	0.2300	~ 17.97M

From Table 1, it is shown that the advanced model confirms a practical approach across all key metrics, SSIM, PSNR, LPIPS, and MAE, due to combining intelligent

preprocessing and a superior architecture, where PCA allows the model to learn from potent and less complex signals. At the same time, the U-Net with skip connections retains the high-resolution spatial information, which is necessary for more accurate and perceptually superior reconstruction.

5.2 Qualitative Analysis

The qualitative analysis shows a different divergence in the two methodologies' performance, visually confirms the quantitative results, and the significant impact of our chosen strategies is highlighted. The Advanced model (with PCA) (Fig. 2(A)) successfully learned to reconstruct meaningful images from the fMRI signals. Figure 2 shows that, even with low resolution and sparse detail, the outputs successfully represent the main structural and spatial features of the ground truth images, like the vertical panes of a stained-glass window and the outline of an animal.

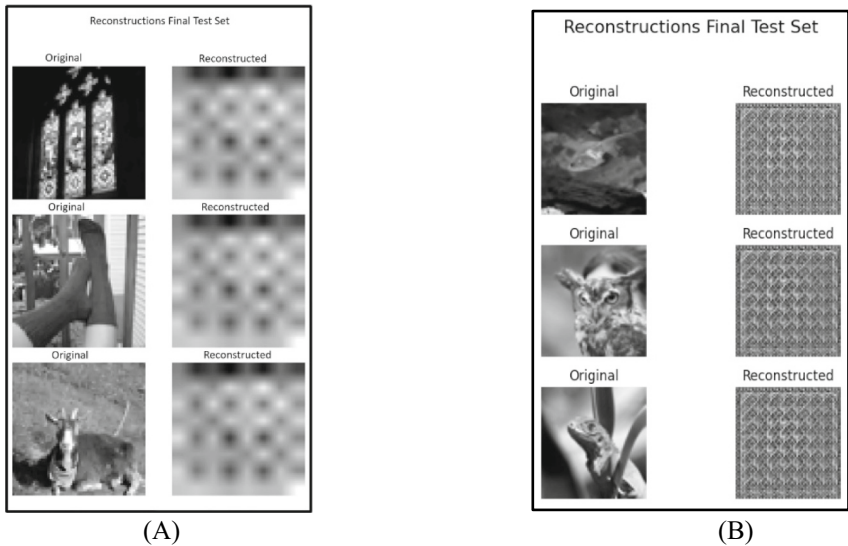


Fig. 2. Qualitative results from the models. This shows that the model learned to map pre-processed neural data into image space conditionally accurately.

By sharp contrast, the Baseline model (no PCA) demonstrated a total breakdown of conditional generation called mode collapse (Fig. 2(B)). Regardless of the fMRI input, the model generated an identical generic, patterned output for each sample. This shows that it failed to learn helpful information from the raw voxel input and instead fell back to a trivial solution.

Their underlying training dynamics account for this extreme contrast in results. The successful reconstructions of the Advanced model were produced from a checkpoint saved through early stopping, successfully avoiding the familiar pattern of overfitting in its loss curve (Fig. 3). In contrast, the Baseline model's deceptively stable loss curve



Fig. 3. Training and validation loss for the advanced model (with PCA), showing overfitting and justifying early stopping.

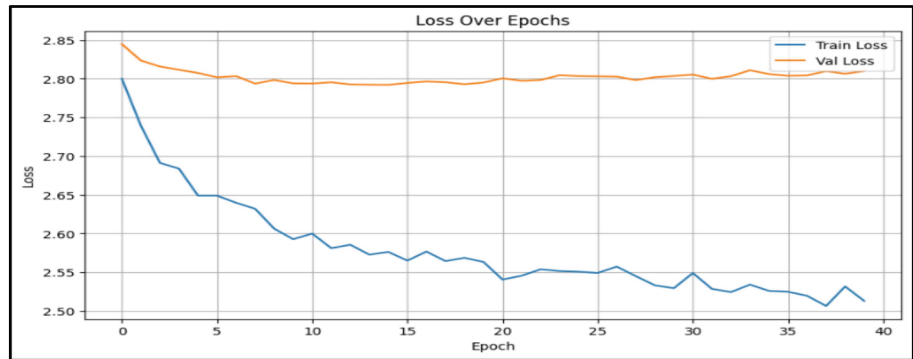


Fig. 4. Training and validation loss for the baseline model. The stable curve reflects convergence to a trivial solution, not successful generalization.

(Fig. 4) did not indicate successful generalization, but rather its rapid convergence to the trivial, low-error, patterned output. The qualitative findings demonstrate that the Advanced model’s hybrid method is successful. In contrast, the purely architectural approach of the Baseline is insufficient to overcome the complexity of the raw fMRI data.

5.3 Comparative Analysis

Table 2 situates our lightweight models within this context, showing their outstanding efficiency. This chart compares reconstruction quality and cost. Our light models with ~ 0.24 SSIM score and fewer parameters and training are better than Diffusion and GAN-based models, which use over 50M parameters and extensive training. The feasibility of cost-effective brain decoding and a key performance-per-cost baseline has been indicated.

Table 2. Comparative analysis of lightweight models versus state-of-the-art

Model	Architectural type	Performance (SSIM)	Model complexity (trainable parameters)	Est. training cost & hardware
MindEye [16]	Heavyweight (diffusion + CLIP)	Very high (92–93% top-1 accuracy; SSIM ~ 0.32–0.48)	Very high (> 600M)	Very high (single NVIDIA A100 GPU,
GAN-based [4]	Heavyweight (GAN + VGG)	High (high Pearson correlation; ~ 70–97% human judgment)	High (> 50M)	High (Caffe-based pre-trained models; CPU/GPU compatible)
Our advanced (with PCA)	Lightweight (U-Net)	~ 0.24	Low (~ 17.97M)	Extremely low (~ 4 min)
Our baseline (w/o PCA)	Lightweight (simple decoder)	~ 0.24	Very low (~ 11.26M)	Extremely low (~ 2 min)

6 Discussion

Light convolution neural networks’ adequate reconstruction and clever preprocessing for visual images influence performance, which is a substantial part of our research, but compared to the baseline model, it collapses. The PCA-based advanced model interprets and gives structured reconstructions (SSIM $\approx$ 0.24). The presence of limited resources, our binary result opens an unequivocal indication of the importance of hybrid solutions, and the divergent behavior of both models sheds the serious multidimensional neural signal deciphering problem. Bagging down the thousands of noisy voxel features, the failure of the baseline model highlights the curse of dimensionality and collapse on a trivial low error solution, but detail has been preserved in the presence of skip connections along a low-dimensional signal by effective de-noising, feature extraction process, and PCA, which translates the coherent images reliably in the U-Net architecture. A critical comparison at the niche level in the context of state-of-the-art has been provided, though our advanced model isn’t comparable with the heavyweight GANs and diffusion models in terms of similarity reconstruction, but it’s minutes of training on a single GPU enables mind-boggling results in computational expanse, along with significant reconstruction comparatively much less than a high-end cluster’s runtime. Our study explores a crucial performance-per-cost benchmark instead of overtaking peak performance, showing tremendous capability for resource-effective methods. Single subject, Low-resolution reconstructions are the limitations, but our structural validity enables future research verifying on multiple subjects and detailing without undermining the

essence of the lightweight philosophy. PCA's optimality power opens a fruitful research direction.

7 Conclusion

We have successfully provided the worth and practicability of a lightweight CNN's reconstruction from fMRI. Our hybrid strategy utilizes intelligent data preprocessing and a strong neural architecture. Though lightweight, it achieves impressive outcomes using only a fraction of the computational resources that are usually necessary. Opening up brain decoding research sets a critical standard for the next generation's performance for more people, where a comparison is made with more powerful models. Despite low fidelity as a limitation, our research establishes clinical feasibility by providing accessibility and speed, and creating functional biomarkers. It makes a reliable and efficient link between neural signals and visual structure, leading to the potential for practical applications like stroke rehabilitation using real-time neurofeedback or communication aids for "locked-in" patients, where speed and low cost are vastly more vital than photorealism. Future work will focus on Multiple subject validation, privacy-preserving techniques like federated learning, and enhancing reconstruction using more advanced, efficient state space models (e.g., Mamba). It proves the practical, insightful achievements of clinically relevant findings using resource-efficient methodologies.

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